

DETERMINATION OF WEIBULL DISTRIBUTION PARAMETERS FROM THREE-POINT FLEXURAL TEST



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Introduction

The inherent variability in ceramic failure stress due to variation in flaw size and distribution prompts a crucial question: Can load-bearing ceramic components still be designed effectively despite this variability? Theoretically, with complete flaw characterization and stress concentration calculations, determining the precise failure stress of a component would be possible based on its flaws and applied stresses, given the material's fracture toughness (K_{IC}). However, this approach is impractical due to the complexities of flaw characterization and the extensive effort involved.

An alternative method involves statistically characterizing the behaviour of numerous samples of the same material. This statistical approach acknowledges the impracticality of individually characterizing flaws of each part and emphasizes designing based on the probability of survival under stress. Designing with brittle solids then becomes a matter of estimating the probability of a part's survival at a given stress level, incorporating parameters derived from the statistical distribution. Engineers must evaluate an acceptable risk factor and determine appropriate design stress based on these considerations.

The two-parameter Weibull distribution serves as a pivotal framework in ceramic design, primarily due to its capacity to effectively capture the inherent variability in material strength arising from internal flaws and defects. This statistical model provides engineers with a method to evaluate the likelihood of failure across diverse stress conditions. Consequently, it offers a practical avenue for ceramic design by enabling the estimation of the probability of component survival at designated stress thresholds. The probability of failure, P_f , at a given stress σ is given by:

$$P_f = 1 - e^{(-\frac{\sigma}{\sigma_0})^m}$$

Where,

m = Weibull modulus

σ_0 = the normalizing stress

Experimental Procedure

- We formed a group of 5 people
- We consulted with the provided standard (ASTM C 67-08) and performed a three-point flexural test to calculate the modulus of rupture of the supplied specimens.
- We calculated the Weibull modulus and normalizing stress using all groups' data by applying linear regression.
- We calculated the ratio of flexural to tensile strength.

Theoretical procedure

MOR determination:

$$S = 3W(1/2-x)/bd^2$$

Where,

S = Modulus of rupture of the specimen at the plane of failure, lb/in.²

W = maximum load indicated by the testing machine, lbf

l = Distance between the supports, in.

b = Net width of the specimen at the plane of failure, in.

d = Depth of the specimen at the plane of failure, in.

x = Average distance from the midspan of the specimen to the plane of failure measured in the direction of the span along the centerline of the bed surface subjected to tension, in.

Flexural to tensile strength ratio determination:

$$\frac{\sigma(3\text{-point bend})}{\sigma(tension)}[2(m+1)^2]^{1/m}$$

Data Table

Table 1: Previous data of our sample & calculated table for further operations

Number of Ordered Data	Measured Strength σ (Mpa)	F	$\ln[-\ln(1 - F)]$	$\ln(\sigma)$
1	0.081382	0.038461538	-3.27702604	-2.5086
2	0.081714	0.076923077	-2.56445594	-2.50453
3	0.1	0.115384615	-2.13891102	-2.30259
4	0.1	0.153846154	-1.83044219	-2.30259
5	0.811	0.192307692	-1.58574866	-0.20949
6	1.55	0.230769231	-1.38105042	0.438255
7	1.6749	0.269230769	-1.20362422	0.515753
8	1.71	0.307692308	-1.04583611	0.536493
9	1.72	0.346153846	-0.90272045	0.542324
10	1.72	0.384615385	-0.77084168	0.542324
11	1.7273	0.423076923	-0.64769946	0.546559
12	2.153	0.461538462	-0.53139121	0.766862
13	2.1914	0.5	-0.42040687	0.784541
14	2.305	0.538461538	-0.31349571	0.835081
15	2.55	0.576923077	-0.20957327	0.936093

16	2.6046	0.615384615	-0.10765025	0.957279
17	2.95	0.653846154	-0.00677109	1.081805
18	3.08	0.692307692	-0.09404782	1.12493
19	3.606	0.730769231	0.195891833	1.282599
20	3.69	0.769230769	0.300050307	1.305626
21	3.69	0.807692308	0.408179685	1.305626
22	4.1584	0.846153846	0.522595459	1.42513
23	4.63	0.884615385	0.646863687	1.532557
24	4.63	0.923076923	0.787195008	1.532557
25	4.869	0.961538462	0.956545405	1.582889

Calculations and Results

Using the data from **the $\ln[-\ln(1 - F)]$** and **$\ln(\sigma)$** columns, taking them on the y-and x-axes, respectively, I can input them into a graph and do linear regression.

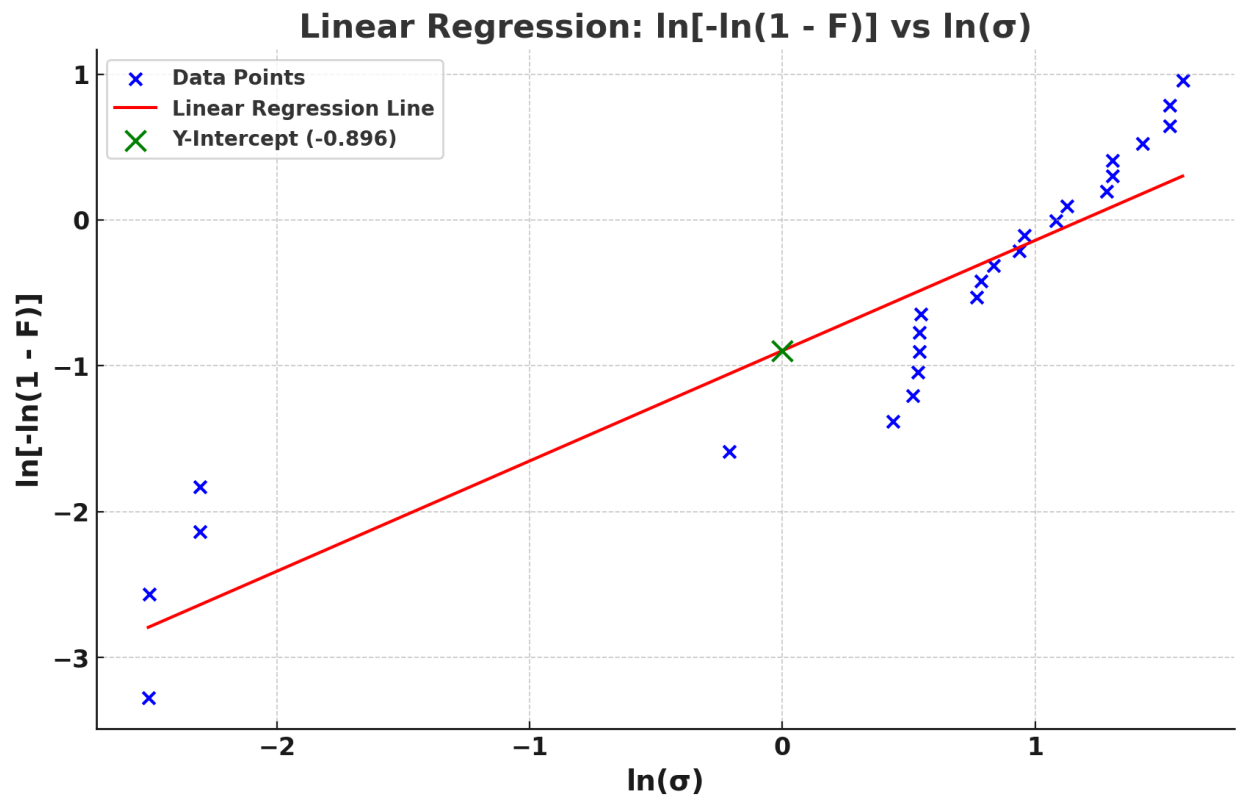


Figure 1: Linear Regression curve to find m & σ_0

From software, the slope of the curve is approximately $m=0.759$. Here $c= -0.896$

We know that, $c= -m\ln\sigma_0$. So, $\sigma_0 = 3.256$ MPa

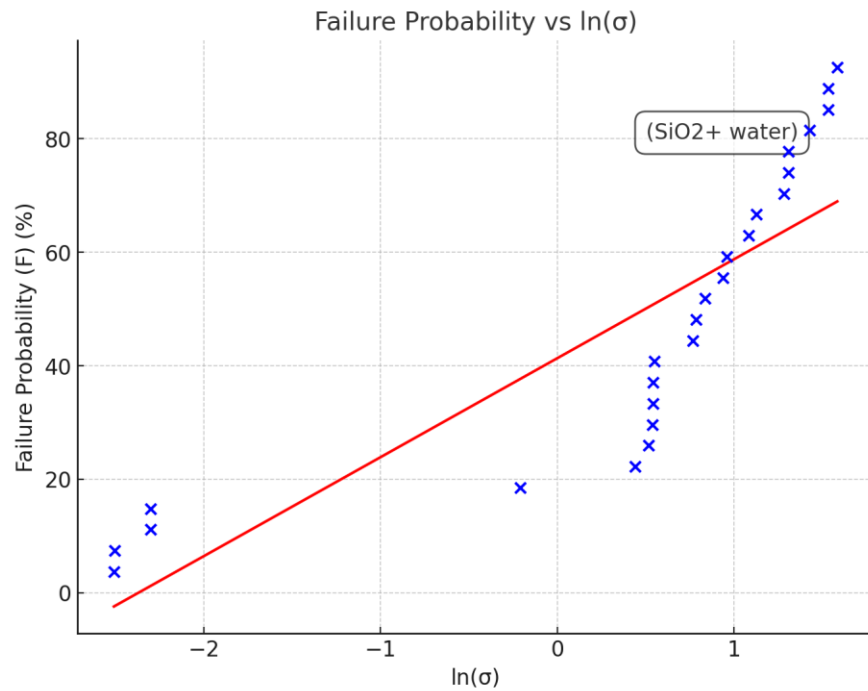


Figure 2: Significance of m value

From this, regression curve we can say it is [bimodal distribution](#), which will be discussed in the discussion section.

Conversion between test:

Measured Strength (σ_{3-pt} [MPa])	Calculated σ_{TS} [MPa]
0.081382	0.007372718
0.081714	0.007402795
0.1	0.009059396
0.1	0.009059396
0.811	0.073471701
1.55	0.140420638
1.6749	0.151735823
1.71	0.154915671
1.72	0.155821611
1.72	0.155821611
1.7273	0.156482947
2.153	0.195048796
2.1914	0.198527604

2.305	0.208819077
2.55	0.231014598
2.6046	0.235961028
2.95	0.267252182
3.08	0.279029396
3.606	0.326681819
3.69	0.334291712
3.69	0.334291712
4.1584	0.376725923
4.63	0.419450034
4.63	0.419450034
4.869	0.44110199

Table 2: Calculated tensile strength from measured strength

Using this formula I calculated σ_{TS} from σ . The value of m was calculated earlier .(m=0.759).

Here, average value of $\sigma = 2.335$ and $\sigma_{TS} = 0.212$

It will be discussed further in the [discussion section](#).

$$\sigma_{TS} = \frac{\sigma_{3\text{-pt}}}{[2(m+1)^2]^{1/m}}$$

Discussion:

Bimodal distribution:

From [Figure: 2](#), we can plot two straight lines in a single curve. Hence, it is a bimodal distribution. One assumption I can make this occurred for two different flaws, like internal flaws & surface flaws. For further applications, we need to convert the bimodal distribution to a monomodal distribution.

Assumption of tensile strength:

From [Table: 2](#), we calculated a tensile strength value for our ceramic materials (Bricks). This is certainly useful to engineers, who do not have proper conditions to measure the tensile strength of all pieces, and it is also quite tough to measure tensile strength after a bend test, because the sample would be broken/ damaged. So, Weibull-based approximation gives a quite accurate assumption of tensile strength after dealing with bend strength.

Question and Answers:

- 1) Coefficient of determination of my regression model is 0.8808
- 2) Some factors are involved to create such variations:
 - While calculating $F \sim n/(N+1)$; this is an approximate value. In real case this can bring some errors.
 - For ceramic materials like ours (bricks), there is no solid margin of errors; if there are more samples, the curve could go further left/ right. So assumption of proper strength in ceramic materials become difficult. In addition frequency vs. range of strength curve for ceramic material is broad, indicates it has large strength scatter. In addition the peak of the curve does not correspond to the average strength and the curve is asymmetrical.

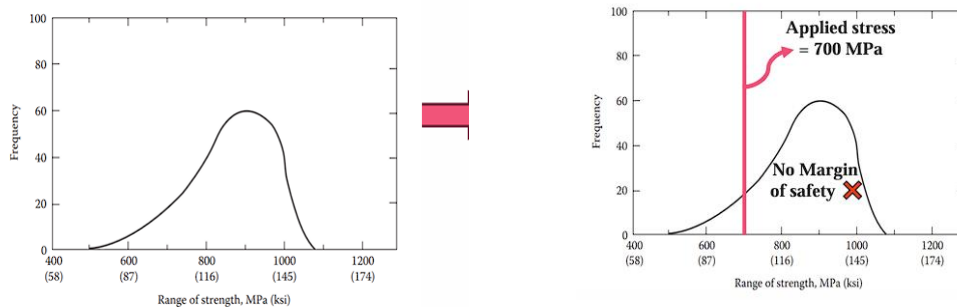


Figure 3: Typical strength distribution curve for ceramic

- Moreover, Weibull probabilistic model is based on “weakest link theory” which predicts that a single flaw is enough to cause failure and also failure will occur at there, which is not true always. Assumption of uniform stress throughout the material also not appropriate all the time.
- 3) Ceramic materials are basically brittle. In bending, microcracks inside them cannot propagate because the compression on one side reduces the effect of the tension on the other side. Ceramics are generally stronger in compression and weaker in tension. Tension causes uniform stress throughout the body, and the presence of even one tiny crack can lead to devastating failure.
 - 4) From our previous lab report,

We know, $\sigma \propto 1/d^2$; when the specimen size is increased to twice σ_{av} becomes 2.335 to 0.3046 MPa (letting all the parameters are constant)

So, $\ln(-\ln(1-F)) = m \ln(\sigma) - m \ln(\sigma_0)$; Given that $F = 0.95$

So, $m = -0.463$

Using the law,

$$\begin{aligned}\sigma_{TS} &= \sigma_0 / [2(m+1)^2]^{1/m} \\ &= 0.09273 \text{ MPa}\end{aligned}$$

Maximum allowable tensile strength 0.09273 MPa

Conclusion

This study validated the application of the Weibull distribution in assessing the failure strength of ceramic materials from the three-point flexural test. The calculated Weibull modulus (m) of 0.759 and normalizing stress (σ_0) of 3.256 MPa confirmed that the material possesses a high level of variability in strength, which is a critical parameter in its design and application. The results of this study indicate that the probabilistic approach effectively approximates the probability of failure, particularly where tensile strength cannot be readily measured.

The bimodal distribution observed here suggests two types of flaws, possibly surface flaws and internal defects. The existing research may well investigate how this bimodal distribution might be converted into a monomodal distribution to be able to have the more precise material design. Lastly, the hypothesis of tensile strength as a function of flexural strength, calculated from the Weibull model, provides engineers with a useful way to estimate material response in real applications, though at the expense of flaw characterization.

Overall, the findings emphasize the importance of statistical methods in material design, particularly in brittle materials like ceramics, where the traditional test methods are hindered by experimental constraints. Weibull statistics form a credible foundation in the designing of ceramic parts to account for material variability inherent to the material to achieve safe and reliable operation under load conditions.